

A Finite-Row Threshold Theorem for $\text{RS}[\mathbb{F}_{17^{32}}, H, 256]$

RS-MCA experimental note

2026-06-27

Abstract

This note packages the high-agreement tangent staircase as a clean finite-row threshold theorem for

$$C = \text{RS}[\mathbb{F}_{17^{32}}, H, 256], \quad |H| = 512.$$

Under the finite-slope support-wise MCA convention used in the experimental ledger, the target $\varepsilon_* = 2^{-128}$ threshold is pinned exactly on the integer grid: integer Hamming radius 5 is safe and integer Hamming radius 6 is already unsafe. Equivalently, with closed Hamming balls, the real safe interval is $[0, 6/512)$; the supremal transition radius is $6/512 = 3/256$, but the endpoint is unsafe.

1 Scope and Definition Audit

The public Proximity Prize page asks, for smooth-domain Reed–Solomon codes at rates

$$\rho \in \{1/2, 1/4, 1/8, 1/16\},$$

to determine the largest radius δ_C^* for which the MCA error is at most $\varepsilon_* = 2^{-128}$ [1]. The row treated here is the rate-1/2 smooth subgroup row

$$F = \mathbb{F}_{17^{32}}, \quad H \leq F^*, \quad |H| = 512, \quad k = 256.$$

The divisibility $512 \mid 17^{32} - 1$ ensures such a multiplicative subgroup exists; it is a power-of-two smooth domain.

This note proves a theorem for the finite-slope support-wise MCA normalization used in the experimental ledger. The exact object is:

- slopes are sampled from the finite line field F , so the denominator is $q_{\text{line}} = |F| = 17^{32}$;
- a closed integer Hamming radius r means agreement threshold $a = n - r$;
- a real closed radius δ means integer radius $r(\delta) = \lfloor \delta n \rfloor$, hence agreement threshold $a(\delta) = n - r(\delta)$;
- support-wise noncontainment is tested on the same support on which the line word is explained by a codeword.

The high-agreement note also proves the matching projective-slope and no-loss CA numerators in the same range. If the sampler is instead uniform on $\mathbb{P}^1(F)$, the numerator below is unchanged in this row and the denominator is $|F| + 1$. The same 5/6 integer-radius threshold remains true because

$$6 \cdot 2^{128} < 17^{32} < 17^{32} + 1 < 7 \cdot 2^{128}.$$

Remark 1 (audit status). The public prize page is a high-level challenge statement; companion paper definitions control the official MCA sampler, finite/projective slope choice, and endpoint convention. This note therefore records an exact local theorem and the definition match that has been checked inside this repository. It is not a peer-reviewed prize submission.

2 Finite-Slope Threshold

Let $D \subset F$, $|D| = n$, and $C = \text{RS}[F, D, k]$. For an agreement threshold a , let

$$B_{\text{fin}}(C, a) = \text{LD}_{\text{sw}}(C, a)$$

be the largest possible number of finite slopes on a received affine line $f + zg$ that admit a codeword explanation on some support of size at least a , but for which f and g are not separately explained on that same support. The finite-slope MCA error at grid agreement a is then

$$\varepsilon_{\text{mca}}^{\text{fin}}(C, a) = \frac{B_{\text{fin}}(C, a)}{|F|}.$$

Lemma 1 (high-agreement tangent staircase). *For $C = \text{RS}[F, D, k]$, $|D| = n$, if*

$$3a - 2n \geq k,$$

then

$$B_{\text{fin}}(C, a) = n - a + 1.$$

For every $k + 1 \leq a \leq n$, the lower bound

$$B_{\text{fin}}(C, a) \geq n - a + 1$$

also holds.

Proof. This is the moving-root tangent floor and the very-high-agreement tangent staircase from `experimental/notes/high_agreement/tangent_staircase.tex`. The proof is finite and uses the MDS rigidity of Reed–Solomon codes plus a residual-root budget. $\square$

Theorem 1 (finite-row MCA threshold). *Let $H \subset \mathbb{F}_{17^{32}}^*$ have order 512, and put*

$$C = \text{RS}[\mathbb{F}_{17^{32}}, H, 256].$$

Under the finite-slope support-wise MCA convention,

$$\varepsilon_{\text{mca}}^{\text{fin}}(C, a) \leq 2^{-128} \quad \text{if and only if} \quad a \geq 507.$$

Here a ranges over integer agreement thresholds $0 \leq a \leq 512$. Equivalently, on the integer Hamming grid, the largest safe radius is

$$r = 5, \quad \delta = 5/512,$$

and the first unsafe radius is

$$r = 6, \quad \delta = 6/512 = 3/256.$$

Proof. Here $n = 512$, $k = 256$, and the high-agreement condition

$$3a - 2n \geq k$$

is $3a - 1024 \geq 256$, namely $a \geq 427$. Lemma 1 therefore gives

$$B_{\text{fin}}(C, a) = 513 - a \quad (a \geq 427).$$

In particular,

$$B_{\text{fin}}(C, 506) = 7, \quad B_{\text{fin}}(C, 507) = 6.$$

The exact field arithmetic is

$$17^{32} = 2367911594760467245844106297320951247361,$$

$$17^{32} - 6 \cdot 2^{128} = 326217393234836465063858652730341978625 > 0,$$

and

$$7 \cdot 2^{128} - 17^{32} = 14064973686101998399515954701426232831 > 0.$$

Thus $6/17^{32} < 2^{-128} < 7/17^{32}$.

For every $a \geq 507$, the exact high-agreement formula gives $B_{\text{fin}}(C, a) \leq 6$, hence the row is safe. At $a = 506$, the formula gives seven bad finite slopes, hence the row is unsafe. For smaller a , closed balls only get larger, so the MCA error cannot decrease; those thresholds are unsafe as well. This proves the claimed grid transition. $\square$

Corollary 1 (closed endpoint language). *With real closed Hamming balls normalized by*

$$r(\delta) = \lfloor 512\delta \rfloor,$$

the finite-slope support-wise MCA safe set is

$$0 \leq \delta < 6/512.$$

The supremal safe radius is

$$\sup\{\delta : \varepsilon_{\text{mca}}^{\text{fin}}(C, \delta) \leq 2^{-128}\} = 6/512 = 3/256,$$

but the endpoint $\delta = 3/256$ is unsafe.

Remark 2 (maximum versus supremum). If a challenge statement asks for a maximum radius satisfying the inequality at the endpoint, the finite real-radius maximum is not attained under this closed endpoint convention; the largest safe grid point is $5/512$. If the intended object is a supremal threshold, the transition value is $3/256$, with the endpoint on the unsafe side.

3 Row-Independent Compiler Target

The preceding row is an instance of a more general finite-grid compiler. Let

$$R = n - k, \quad r = n - a, \quad B_Q = \left\lfloor \frac{Q}{2^{128}} \right\rfloor,$$

where Q is the denominator used by the sampler or protocol term.

Theorem 2 (high-agreement threshold compiler). *Assume $C = \text{RS}[F, D, k]$, $|D| = n$, and that the relevant line/MCA/CA object is in the high-agreement tangent regime. If*

$$1 \leq B_Q \leq \left\lfloor \frac{n - k}{3} \right\rfloor,$$

then the single line/MCA/CA grid threshold at target 2^{-128} is pinned exactly:

$$r \leq B_Q - 1 \implies \text{safe},$$

while

$$r = B_Q \implies \text{unsafe}.$$

Under the same closed-radius convention as Corollary 1, the safe real interval is $[0, B_Q/n)$, the supremal transition radius is B_Q/n , and the endpoint is unsafe.

Proof. For $r \leq B_Q - 1$, the high-agreement condition is valid because

$$r \leq B_Q - 1 < B_Q \leq \left\lfloor \frac{n - k}{3} \right\rfloor.$$

The tangent staircase gives numerator $r + 1 \leq B_Q$. By definition of $B_Q = \lfloor Q/2^{128} \rfloor$, this gives

$$\frac{r + 1}{Q} \leq 2^{-128}.$$

At $r = B_Q$, the same exact formula gives numerator $B_Q + 1$, and

$$B_Q + 1 > \frac{Q}{2^{128}},$$

so the error is strictly larger than 2^{-128} . Larger radii are unsafe by monotonicity of closed balls. $\square$

For the row of Theorem 1,

$$Q = 17^{32}, \quad B_Q = 6, \quad \left\lfloor \frac{512 - 256}{3} \right\rfloor = 85,$$

so the compiler applies with substantial room.

4 Scanner Attachment

The reproducibility attachment is generated from the repository root by running `python3` on `experimental/notes/certificate_scanner/certificate_scanner.py` with input `experimental/notes/certificate_scanner/examples/f17_512_mca_only.json`; the JSON and Markdown outputs are written to `experimental/notes/certificate_scanner/outputs/f17_512_mca_only.report.json` and `experimental/notes/certificate_scanner/outputs/f17_512_mca_only.report.md`. The attached report confirms:

$$\left\lfloor \frac{17^{32}}{2^{128}} \right\rfloor = 6,$$

and gives the following pure line/MCA verdicts.

a	$\sigma = a - k$	$r = n - a$	exact line numerator	verdict
506	250	6	7	unsafe
507	251	5	6	safe
508	252	4	5	safe
512	256	0	1	safe

The older combined protocol scanner profile also includes an interleaved-list term. That is a different ledger: with one extra list term over the same denominator, the numerator at $a = 507$ is $6 + 1$, so the combined line-plus-list ledger is unsafe at $a = 507$ and safe at $a = 508$. The theorem in this note is the pure finite-slope support-wise MCA threshold.

References

- [1] The Proximity Prize. Public challenge page. <https://proximityprize.org/>, accessed 2026-06-27.